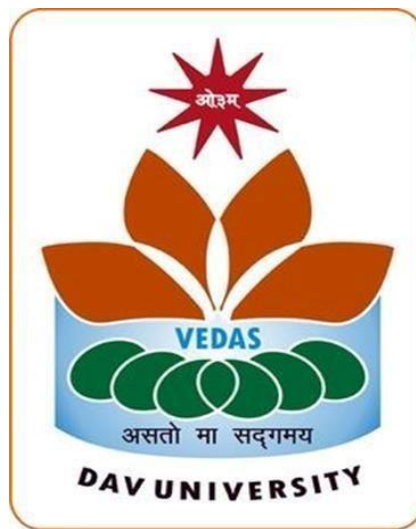


Project Report
On
TRAVEL MANAGEMENT
Submitted in the partial fulfilment of the requirement for the award of degree
of
Bachelor of Technology
In
Computer Science and Engineering
Batch (2022-2026)



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DECLARATION

I, Mehul katoch declare that the work presented in this project report titled “**Travel management**” submitted in partial fulfillment of the requirements for the award of Bachelor of Technology (B.Tech) in Computer Science and Artificial Intelligence is an authentic record of my own work carried out under the guidance of **Mr. Sanjeev Dhiman**

To the best of my knowledge, the matter embodied in this report has not been submitted to any other University/ Institute for the award of any degree or diploma.

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ABSTRACT

The recent past showed a greater interest in recommender techniques. Now-a-days there are many travel packages existing from different websites to almost all the places over the world. A customer finds it very difficult to search for the best package as he/she has to browse multiple websites, contact many travel agents and etc. which is a tedious process and is time consuming. There should be a system where the user should find the best package on the Internet with a single click. To address this issue, we adopt Travel Package Recommendation System which offers the best package among all the other packages that are on the web. This project will help tourist to suggest the best Travel Package among all the package deals on the web. On multiple demands of tourist that is, a customer will select a travel package for a particular place based on the recommendations provided by the previous customers who had experience with the package. Therefore, according to the personalized recommendations, he/she will choose the best package that is on the web. Initially, we will evaluate the particular characteristics of the current traveling packages and we mine the data on the tourists rating and the intrinsic features i.e., locations, travel seasons etc. Based on the data collected after mining, we will generate a list for personalized travel package recommendations. Furthermore, we will extract the data based on the tourist's relation with the area and season.

TABLE OF CONTENTS

SR. NO.	CONTENTS	PAGE NO.
1.	Introduction to Project 1.1 Purpose and significance 1.2 Objectives 1.3 Module 1.3.1 Module Description	9-13
2.	Existing System 2.1 Drawback	14
3.	Proposed System 3.1 System overview 3.2 Features of proposed system 3.3 Benefits Of Proposed System 3.4 Expected outcome	15-17
4.	Feasibility Study 4.1 Feasibility study	18-19
5.	Software Requirement Specification 5.1 Definition 5.2 Requirements	20-21

6.	Technology Used 6.1 Technologies Used 6.2 Tools and Editors Used 6.2.1 VS code	22-31
7.	Implementation Results	32-36
8.	System Testing 8.1 Introduction to Testing	37-39
9.	Conclusion and Future Scope 9.1 Conclusion 9.2 Future Scope	40-41
10.	References	42

LIST OF FIGURES

SR. NO.	FIGURE NAME	PAGE NO.
1.	Fig 1.1:Flowchart of package module	12
2.	Fig 1.2: transportation module	13
3.	Fig 6.1: HTML logo	22
4.	Fig 6.2: CSS logo	23
5.	Fig 6.3: MYSQL logo	24
6.	Fig 6.4: JS logo	25
7.	Fig 6.5: Php logo	27
8.	Fig 6.6: UML logo	28
9.	Fig 6.7: VS Code Studio	30

CHAPTER 1

INTRODUCTION TO PROJECT

1.1 Purpose and significance

The Tours and Travel Management System is a web based application. The main purpose of “Tours and Travels Management System” is to provide a convenient way for a customer to book hotels, flight, train and bus for tour purposes. The objective of this project is to develop a system that automates the processes and activities of a travel agency. In this project, we will make an easier task of searching places and for booking train, flight or bus. In the present system a customer has to approach various agencies to find details of places and to book tickets. This often requires a lot of time and effort. We provide approach skills to critically examine how a tourist visits and its ability to operate in an appropriate way when dealing with the consequences of tourism, locally, regionally, and nationally including visitor security and ecological influences. It is tedious for a customer to plan a particular journey and have it executed properly. The project ‘Tours and Travels Management System’ is developed to replace the currently existing system, which helps in keeping records of the customer details of destination as well as payment received.

Tourism can be considered as most favorite pass time when people get free time. Several travel organizations are available on the web. The people or the tourist selects their own Travel Package according to their personal interest. The travel companies concentrate on the interest associated with tourist making sure to increase their particular market value and supply enormous package deals, so that they can make their Travel Package more effective. Now-a-days Recommender system is becoming very famous and people are getting attracted to it, as it is helping them to choose the best package in a short time. With this, item recommendation is analyzed then it retrieves the information and filters this for research. For example, if the tourist goes to hill stations more often, then database contains “hill station” as recommendation rely on the similar factors of user or items. Preferences of different users for same item are recommended by system

1.2 Objectives

Faster processing time and more accurate data for travel requests and reimbursements

- The Travel Packages will be presented based on the interest of the tourist.
- By using tourist, area and season as our inputs we can represent our travel data in the best form.
- By using this recommendation approach the flaws of the existing system will be eliminated as it performs much better than traditional techniques.
- The algorithm ‘Weighted Average Entropy’ will help the tourist to find the best package in the particular area based on season and theme.

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- Ability for travellers to track authorization and reimbursement request status through the system rather than via phone calls or campus mail
- Major technological upgrades to the current travel system
- Use of IU's standardized, virtual J2EE environments
- Many new features and enhancements
- Promote responsible and interesting tourism so that people can enjoy their holidays at their favorable places.
- Provide a search platform where a tourist can find their tour places according to their choices.

1.3 Module

The entire project mainly consists of following modules, which are

- Administrator module
- Hotel module
- Transportation module
- Package module
- Testimonials module
- Search Module
- Routes module
- Reservations module
- Tour Details

1.3.1 Module Description

Administrator module:

This module provides administrator related functionality like from this module use can add route information, bus information, train information, flight information, tour packages, travel packages, bus seat details, etc. From this module Admin can view daily, weekly and monthly report. This module is develop for admin of the website and admin can add, delete, edit and view the data related to places, travels, routes, bookings from this module.

Transportation Module:

a. Flight

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- b. Train
- c. Bus

Package Modules

User can view different tour packages available for tourist. User can select any packages from this module he can also check the details of various travel agencies. A user can select any travel agency from this module.

Testimonials module:

This is testimonials module where passenger can post feedback after the journey and they can share their experience. Users of this application can post their opinions, complaints.

- Photos
- Videos

Routes module:

This will display the route information of Source location and destination location. User can also check bus routes for his destination. User can check bus route train route and car route for his journey and can select any route packages from the available tour packages. From this module user can also get information related to various routes connecting sources and destinations. For each route, information such as source, destination, fare, reservation details, pick up points etc are provided.

Reservations module:

This module is for passengers where passenger can reserve the seats by making payment. Using this module user can book bus ticket or train ticket. From this module user can also book tickets or cancel previously booked tickets. The module maintains the details of all reservations made so far and allows administrator to either confirm or reject the bookings. Accordingly, the administrator can take various steps to act on the complaints and suggestions.

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Tour Details:

Details of different types of tours which include tours like family tours, couple tours, general tours, date and time of departure and the fair of the tours etc are maintained. As the customer ask for the details of a particular tour, the tours and travel management system gives the details of the related place where tourist wants to go and the date and time of the tour, no. of seats available of that particular tour, fair of the tour, details of discount on a particular tour package if any. Details of the pick-up facilities and the drop facilities are also maintained.

In today's rapidly evolving digital landscape, users increasingly expect AI assistants to go beyond simple text-based responses and handle multimodal, vision, and creativity within a single conversation. However, most existing platforms force users to switch between separate tools — one for chatting, another for generating images, and yet another for analyzing pictures — breaking the flow of interaction and losing conversational context.

NovaChat addresses this gap by providing a unified, intelligent chat interface where users can:

- Converse naturally with an AI assistant
- Generate high-quality images directly from their text descriptions
- Upload images and instantly receive meaningful captions and insights

All within the same persistent chat thread, without ever leaving the conversation. This seamless multimodal experience eliminates fragmented workflows, maintains full context across interactions, and delivers a more intuitive, human-like AI companionship — making advanced AI capabilities accessible and practical for everyday use.

Flow chart:

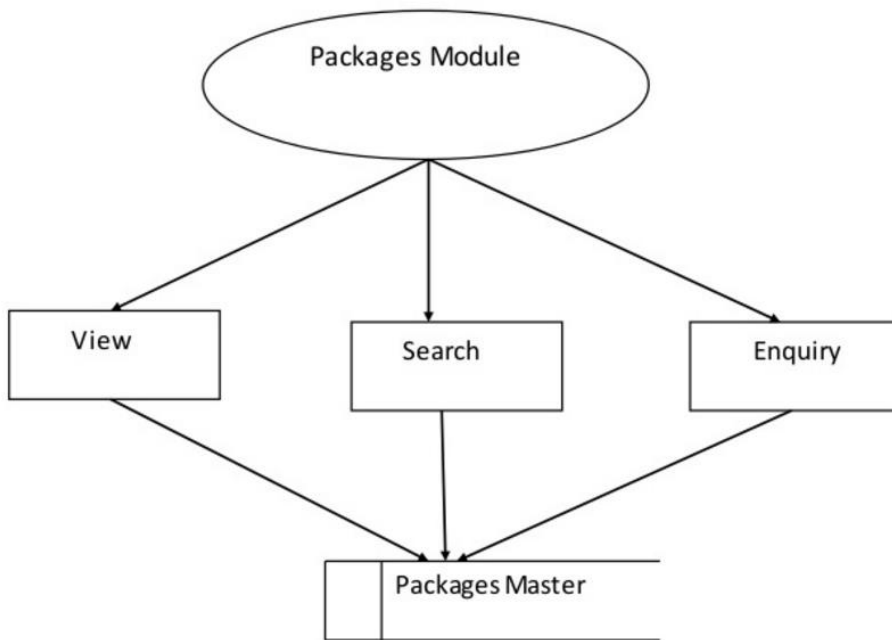


Fig .1.1

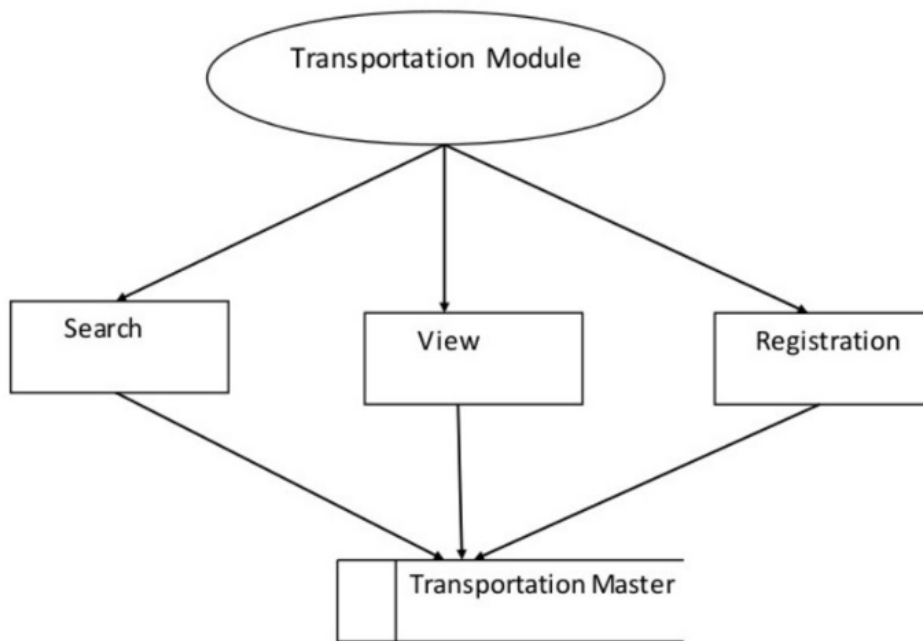


Fig .1.2

CHAPTER 2

EXISTING SYSTEM

Tour and Travels currently use a manual system for the management and maintenance of critical information. The current system requires numerous paper forms, with data stores spread throughout the Tour and Travel management infrastructure. Often information is incomplete or does not follow management standards. Forms are often lost in transit between departments requiring a comprehensive auditing process to ensure that no vital information is lost. Multiple copies of the same information exist in the Tour and Travel and may lead to inconsistencies in data in various data stores.

2. Drawback of existing system

1. **Time-consuming processes:** Manual handling of bookings, records, and customer requests requires significant time and effort, slowing down overall operations.
2. **High risk of data loss:** Paper forms and physical documents can be easily misplaced, damaged, or lost, leading to incomplete or missing information.
3. **Data duplication and inconsistency:** Multiple copies of the same information may exist across different departments, causing confusion and inaccurate decision-making.
4. **Lack of centralized data:** Information is scattered throughout the organization, making it difficult to access or update records efficiently.
5. **Poor accuracy and human error:** Manual data entry increases the chances of typing errors, misinformation, and incorrect bookings.
6. **Difficult to track updates:** Changes in customer details, travel plans, or bookings are not updated consistently, resulting in outdated records.
7. **Inefficient communication:** Coordination between departments is slow due to reliance on physical files and manual communication methods.
8. **Limited customer service efficiency:** Without quick access to accurate data, staff cannot respond promptly to customer queries or provide personalized services.
9. **No real-time information:** The system cannot provide immediate updates on availability, pricing, or travel options, affecting customer satisfaction.
10. **Lack of security:** Paper-based systems offer minimal protection against unauthorized access, damage, or misuse of sensitive information.

CHAPTER 3

PROPOSED SYSTEM

The Tour and Travel Management System is designed for any tourism to replace their existing manual paper based system. The new system is to control the information of customers. Package availability, staff and travel services and so on . These services are to be provided in an efficient, cost effective manner, with the goal of reducing the time and resources currently required for such tasks understanding as natural parts of the same interaction, rather than isolated features.

3.1 System Overview

The proposed Travel Management System is a comprehensive, automated platform designed to simplify and streamline all activities related to planning, booking, and managing travel services. It replaces traditional manual methods—such as physical bookings, phone-based inquiries, and spreadsheet-based planning—with a centralized digital system that is faster, more accurate, and easier to operate.

The system integrates various travel operations including destination search, itinerary planning, booking of transport and hotels, cost estimation, payment processing, and user profile management. Users can explore destinations, compare prices, customize travel packages, and make secure reservations through an intuitive interface.

In the backend, the system maintains a centralized database that stores user details, booking records, travel packages, pricing, and transaction logs. The system ensures real-time updates, eliminating duplicate bookings and reducing human errors. Additionally, automated notifications keep users informed about booking status, offers, and travel reminders.

The proposed system enhances user experience through features like intelligent recommendations, dynamic pricing updates, automated itinerary generation, and secure payment gateways. It is designed to support multiple user roles such as customers, travel agents, and system administrators.

Overall, the Travel Management System provides a complete, integrated, and user-friendly platform that improves efficiency, accuracy, and convenience in the travel planning and management process.

3.2 Features Of Proposed System

1. **To automate the entire travel management process:** The main objective is to replace manual paperwork with a fully automated system that streamlines booking, customer management, package handling, and staff coordination, reducing human effort and operational delays.

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2. **To provide a centralized database of all travel-related information:**The system aims to store customer data, travel packages, bookings, payments, and staff details in a single, unified database, ensuring easy access, consistency, and accuracy of information across all departments.
3. **To improve the efficiency and speed of travel operations:**The system is designed to minimize repetitive tasks such as manual data entry, searching for records, or verifying package availability, thus speeding up the entire workflow and improving productivity.
4. **To reduce errors and inconsistencies caused by manual processes:**By automating data input and validation, the system significantly lowers the risk of human errors, incomplete records, duplication, and miscommunication between departments.
5. **To enhance customer experience and satisfaction:**The system provides timely and accurate information about packages, availability, pricing, and booking status, enabling faster responses to customer queries and improving overall service quality.
6. **To enable real-time updates and tracking;**The platform supports instant updates to booking details, package modifications, cancellations, and staff schedules, helping both customers and administrators stay informed at all times.
7. **To improve security and protect sensitive data:**The system ensures that customer information, payment details, and internal records are stored securely with proper access control, reducing the risk of unauthorized access or data loss.
8. **To support better decision-making through reports and analytics:**The system can generate detailed reports on bookings, revenue, customer trends, and staff performance, helping management make informed decisions and plan future strategies effectively.
9. **To reduce operational costs;**By minimizing paperwork, reducing errors, and automating tasks, the system helps lower labor costs, resource usage, and overall administrative expenses.
10. **To enhance coordination among different departments:**All departments—booking, customer service, operations, and management—can access shared and updated information, ensuring smooth communication and consistent workflow.

3.3 Benefits Of Proposed System

1. **Centralized Data Storage:** All information—customer details, bookings, packages, payments, and staff records—is stored in a centralized database. This eliminates duplication, minimizes errors, and ensures consistency across departments.
2. **Enhanced Accuracy:** Automation reduces human errors caused by manual data entry. Real-time validation ensures that information stored in the system is complete, up-to-date, and reliable.
3. **Better Customer Experience:** Customers can easily search for destinations, compare packages, customize trips, and make bookings online. Instant confirmations and automated notifications improve transparency and convenience.

3.4 Outcome

The implementation of the Travel Management System results in a streamlined, efficient, and highly reliable platform for managing all travel-related operations. The system successfully replaces the traditional manual process with a fully automated and user-friendly interface. As a result, data handling becomes faster, more accurate, and more secure.

The system enables customers to easily search for travel packages, view availability, compare options, and make bookings with real-time updates. Travel agencies benefit from reduced workload, improved coordination between departments, and centralized control over information. Administrative tasks such as record maintenance, invoice generation, and reporting become more manageable and error-free.

Overall, the outcome is a modern, scalable, and cost-effective solution that enhances customer satisfaction, improves operational efficiency, strengthens decision-making through analytics, and contributes to the overall growth and professionalism of the travel organization.

CHAPTER 4

FEASIBILITY STUDY

4.1 What Is Feasibility Study

This study is carried out to check the economic impact will have on the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products have to be purchased.

4.1.1 Technical Feasibility

This study is carried out to check the technical feasibility, that is the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes for the implementing this system.

4.1.2 Economic Feasibility

Economic feasibility determines whether the proposed Travel Management System is financially viable and worth the investment when compared to the existing manual system. The analysis shows that the cost of developing, implementing, and maintaining the automated system is justified by the long-term financial benefits it provides. Although initial expenses include software development, hardware setup, staff training, and system integration, these costs are significantly lower than the recurring expenses associated with manual processes. The new system reduces expenses related to paperwork, file storage, printing, and physical record maintenance. It also minimizes losses caused by human errors, inconsistent data, and poor coordination. By automating routine tasks, the system reduces the need for additional administrative staff, which leads to considerable cost savings over time. Additionally, faster customer service, improved booking accuracy, and better package management contribute to higher customer retention and increased revenue. With features like automated invoicing, real-time updates, and integrated payment processing, the system enhances operational efficiency and profitability. Overall, the economic analysis indicates that the investment in the Travel Management System is cost-effective and will provide substantial financial returns, making the project economically feasible for long-term use.

4.1.3 Operational Feasibility

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of

TRAVEL MANAGEMENT

training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 5

SYSTEM REQUIREMENT SPECIFICATION

The purpose of this SRS document is to define the complete functionality, constraints, and interfaces of the proposed Travel Management System. It provides a detailed description of system features, performance requirements, and operational needs to ensure successful design, development, and implementation.

5.1 Definitions

- **TMS** – Travel Management System
- **Admin** – Person managing backend operations
- **User/Customer** – View, search, and book travel packages
- **Staff** – Handles bookings, queries, and customer support

5.2 Hardware and Software Requirements

5.2.1 Hardware Requirements:

The most common set of requirements defined by any operating system or software application is the physical computer resources, also known as hardware. A hardware requirements list is often accompanied by a hardware compatibility list (HCL), especially in case of operating systems. An HCL lists tested, compatibility and sometimes incompatible hardware devices for a particular operating system or application. The following sub-sections discuss the various aspects of hardware requirements.

Hardware Requirements for Present project:

Processor : Intel dual Core i3

Ram : 1 GB

Hard Disk : 80 GB

5.2.2 Software Requirements :

Software Requirements deal with defining software resource requirements and pre-requisites that need to be installed on a computer to provide optimal functioning of an application. These requirements or pre-requisites are generally not included in the software installation package and need to be installed separately before the software is installed.

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Software Requirements for Present Project:

Operating System: Windows 7/ XP/8

Frontend : HTML, CSS, Bootstrap, JQuery, Javascript.

Server and Script: PHP

Database : MySQL

CHAPTER 6

TOOLS AND TECHNOLOGIES USED

6.1 Technologies Used

6.1.1.HTML:



Fig.6.1

HTML or Hypertext Markup Language is the standard markup language used to create web pages.

HTML is written in the form of HTML elements consisting of tags enclosed in angle brackets (like `<html>`). HTML tags most commonly come in pairs like `<h1>` and `</h1>`, although some tags represent empty elements and so are unpaired, for example `<img>`. The first tag in a pair is the start tag, and the second tag is the end tag (they are also called opening tags and closing tags). Though not always necessary, it is best practice to append a slash to tags which are not paired with a closing tag.

The purpose of a web browser is to read HTML documents and compose them into visible or audible web pages. The browser does not display the HTML tags, but uses the tags to interpret the content of the page. HTML describes the structure of a website semantically along with cues for presentation, making it a markup language rather than a programming language.

HTML elements form the building blocks of all websites. HTML allows images and objects to be embedded and can be used to create interactive forms. It provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. It can embed scripts written in languages such as JavaScript which affect the behavior of HTML web pages.

6.1.2 Cascading Stylesheets (CSS):



Fig.6.2

It is a style sheet language used for describing the look and formatting of a document written in a markup language. While most often used to style web pages and interfaces written in HTML and XHTML, the language can be applied to any kind of XML document, including plain XML, SVG and XUL. CSS is a cornerstone specification of the web and almost all web pages use CSS style sheets to describe their presentation.

CSS is designed primarily to enable the separation of document content from document presentation, including elements such as the layout, colors, and fonts. This separation can improve content accessibility, provide more flexibility and control in the specification of presentation characteristics, enable multiple pages to share formatting, and reduce complexity and repetition in the structural content .

CSS can also allow the same markup page to be presented in different styles for different rendering methods, such as on-screen, in print, by voice (when read out by a speech-based browser or screen reader) and on Braille-based, tactile devices. It can also be used to allow the web page to display differently depending on the screen size or device on which it is being viewed. While the author of a document typically links that document to a CSS file, readers can use a different style sheet, perhaps one on their own computer, to override the one the author has specified. However if the author or the reader did not link the document to a specific style sheet the default style of the browser will be applied.

6.1.3 MYSQL



Fig.6.3

MySQL is developed, distributed, and supported by Oracle Corporation. MySQL is a database system used on the web it runs on a server. MySQL is ideal for both small and large applications. It is very fast, reliable, and easy to use. It supports standard SQL. MySQL can be compiled on a number of platforms.

The data in MySQL is stored in tables. A table is a collection of related data, and it consists of columns and rows. Databases are useful when storing information categorically.

Features of MySQL:

Internals and Portability:

- Written in C and C++.
- Tested with a broad range of different compilers.
- Works on many different platforms.
- Tested with Purify (a commercial memory leakage detector) as well as with Val grind, a GPL tool.
- Uses multi-layered server design with independent modules.

Security:

- A privilege and password system that is very flexible and secure, and that enables host-based verification.
- Password security by encryption of all password traffic when you connect to a server.

Scalability and Limits:

- Support for large databases. We use MySQL Server with databases that contain 50 million records. We also know of users who use MySQL Server with 200,000 tables and about 5,000,000,000 rows.

Connectivity:

Clients can connect to MySQL Server using several protocols:

- Clients can connect using TCP/IP sockets on any platform.

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- On Windows systems in the NT family (NT, 2000, XP, 2003, or Vista), clients can connect using named pipes if the server is started with the --enable-named-pipe option. In MySQL 4.1 and higher, Windows servers also support shared-memory connections if started with the --shared-memory option. Clients can connect through shared memory by using the --protocol=memory option.
- On UNIX systems, clients can connect using Unix domain socket files.

Localization:

- The server can provide error messages to clients in many languages.
- All data is saved in the chosen character set.

Why to use MySQL:

- Leading open source RDBMS
- Ease of use – No frills
- Fast
- Robust
- Security
- Multiple OS support
- Free
- Technical support
- Support large database– up to 50 million rows, file size limit up to 8 Million

6.1.4 JavaScript:



Fig.6.4

JavaScript is the scripting language of the Web. All modern HTML pages are using JavaScript. A scripting language is a lightweight programming language. JavaScript code can be inserted into any HTML page, and it can be executed by all types of web browsers. JavaScript is easy to learn.

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JavaScript (JS) is a programming language and core technology of the Web, alongside HTML and CSS. It was created by Brendan Eich in 1995. Ninety-nine percent of websites use JavaScript on the client side for webpage behavior. Web browsers have a dedicated JavaScript engine that executes the client code.

Why to use JavaScript:

JavaScript is one of the 3 languages all web developers must learn:

1. HTML to define the content of web pages
2. CSS to specify the layout of web pages
3. JavaScript to specify the behavior of web pages

Example

```
x = document.getElementById("demo"); //Find the HTML element with id="demo"
x.innerHTML = "Hello JavaScript"; //Change the content of the HTML element
```

document.getElementById() is one of the most commonly used HTML DOM methods.

Other uses of JavaScript:

- Delete HTML elements
- Create new HTML elements
- Copy HTML elements
- In HTML, JavaScript is a sequence of statements that can be executed by the web browser.

JavaScript Statements:

- JavaScript statements are "commands" to the browser.
- The purpose of the statements is to tell the browser what to do.
- This JavaScript statement tells the browser to write "Hello Dolly" inside an HTML element with id="demo":
- Semicolon;
- Semicolon separates JavaScript statements.
- Normally you add a semicolon at the end of each executable statement.
- Using semicolons also makes it possible to write many statements on one line.

JavaScript Code:

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- JavaScript code (or just JavaScript) is a sequence of JavaScript statements.
- Each statement is executed by the browser in the sequence they are written.
- This example will manipulate two HTML elements:

Example

```
document.getElementById("demo").innerHTML="HelloDolly";  
document.getElementById("myDIV").innerHTML="How are you?";
```

JavaScript Properties:

- Properties are the values associated with a JavaScript object.
- A JavaScript object is a collection of unordered properties.
- Properties can usually be changed, added, and deleted, but some are read only.

6.1.5PHP:



Fig.6.5

What is PHP?

- PHP is an acronym for "PHP Hypertext Preprocessor"
- PHP is a widely-used, open source scripting language
- PHP scripts are executed on the server
- PHP costs nothing, it is free to download and use

What is PHP file?

- PHP files can contain text, HTML, CSS, JavaScript, and PHP code
- PHP code are executed on the server, and the result is returned to the browser as plain HTML
- PHP files have extension ".php"

What can PHP do?

- PHP can generate dynamic page content
- PHP can create, open, read, write, delete, and close files on the server
- PHP can collect form data
- PHP can send and receive cookies
- PHP can add, delete, modify data in your database
- PHP can restrict users to access some pages on your website
- PHP can encrypt data

With PHP you are not limited to output HTML. You can output images, PDF files, and even Flash movies. You can also output any text, such as XHTML and XML.

Why PHP?

- PHP runs on various platforms (Windows, Linux, Unix, Mac OS X, etc.)
- PHP is compatible with almost all servers used today (Apache, IIS, etc.)
- PHP supports a wide range of databases
- PHP is free. Download it from the official PHP resource: www.php.net

6.1.6 Introduction to UML:

UML Design

Introduction to UML:



Fig.6.6

The Unified Modeling Language (UML) is a standard language for specifying, visualizing, constructing, and documenting the software system and its components. It is a graphical language, which provides a vocabulary and set of semantics and rules. The UML focuses on the conceptual and physical representation of the system. It captures the decisions and understandings about systems that must be constructed. It is used to understand, design, configure, maintain, and control information about the systems.

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The UML is a language for:

- Visualizing
- Specifying
- Constructing
- Documenting

Visualizing

Through UML we see or visualize an existing system and ultimately we visualize how the system is going to be after implementation. Unless we think, we cannot implement. UML helps to visualize, how the components of the system communicate and interact.

Specifying

Specifying means building, models that are precise, unambiguous and complete UML addresses the specification of all the important analysis design, implementation decisions that must be made in developing and deploying a software system.

Constructing

UML models can be directly connected to a variety of programming language through mapping a model from UML to a programming language like JAVA or C++ or VB. Forward Engineering and Reverse Engineering is possible through UML.

Documenting

The Deliverables of a project apart from coding are some Artifacts, which are critical in controlling, measuring and communicating about a system during its developing requirements, architecture, design, source code, project plans, tests, prototypes, releases, etc...

A diagram is the graphical presentation of a set of elements, most often rendered as a connected graph of vertices and arcs. You draw diagram to visualize a system from different perspective, so a diagram is a projection into a system. For all but most trivial systems, a diagram represents an elided view of the elements that make up a system. The same element may appear in all diagrams, only a few diagrams, or in no diagrams at all. In theory, a diagram may contain any combination of things and relationships. In practice, however, a small number of common combinations arise, which are consistent with the five most useful views that comprise the architecture of a software-intensive system. For this reason, the UML includes nine such diagrams:

- Class diagram
- Object diagram
- Use case diagram
- Sequence diagram

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- Collaboration diagram
- State chart diagram
- Activity diagram
- Component diagram
- Deployment diagram
-

6.2 TOOL USED

6.2.1 VSCODE



Fig. 6.7

Visual Studio Code (VS Code) is a lightweight yet powerful open-source code editor developed by Microsoft. It works on Windows, macOS, Linux, and even web browsers. VS Code supports debugging, syntax highlighting, IntelliSense, code refactoring, Git integration, and a wide range of extensions, making it one of the most popular development tools today.

6.2.1.1 History

- Announced by Microsoft on April 29, 2015 at the Build Conference
- Open-sourced under the MIT License on November 18, 2015
- Officially released on April 14, 2016
- Grew rapidly due to its extension ecosystem and community support

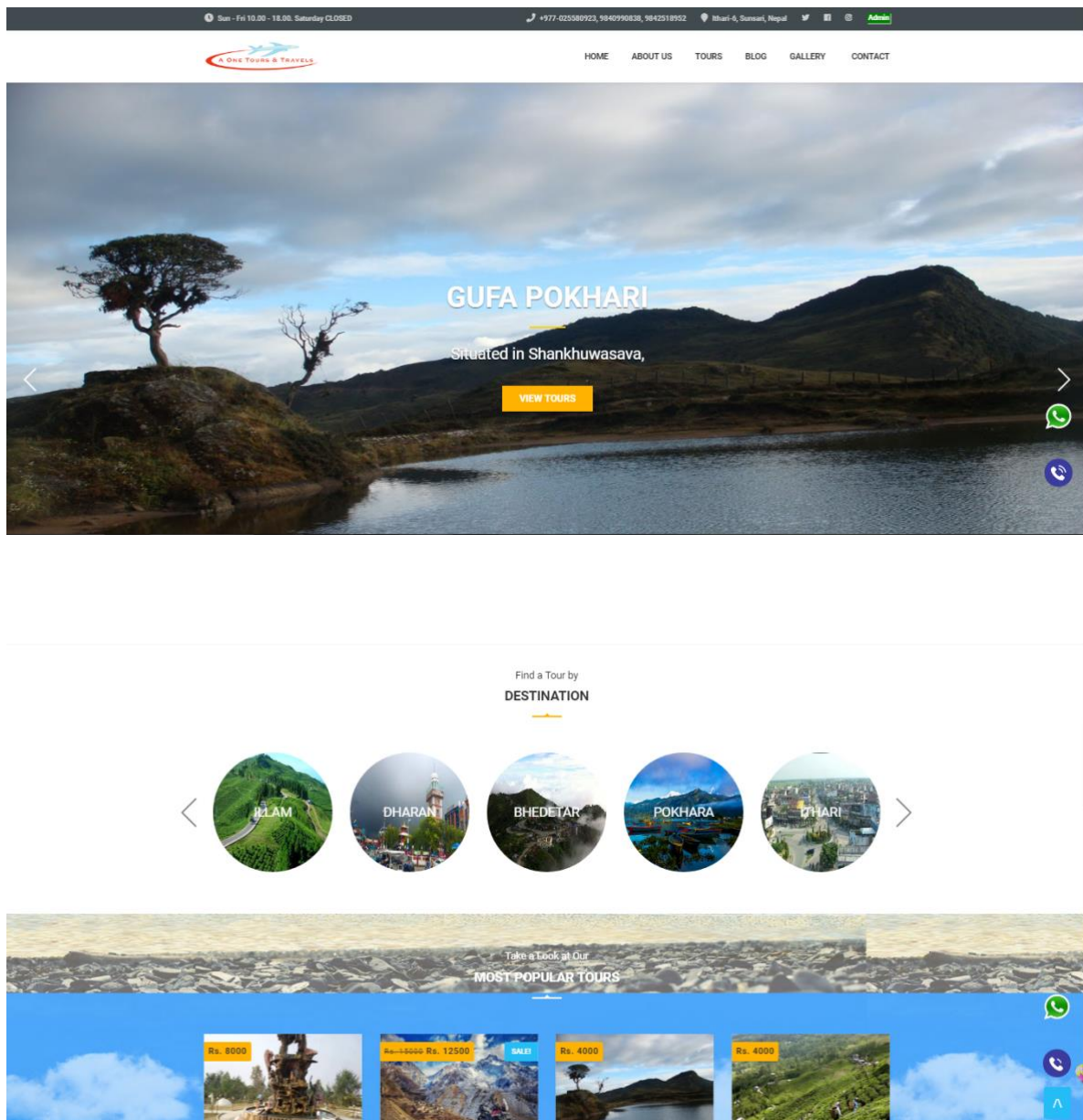
6.2.1.2 Features

- **Code Editing:** Syntax highlighting, auto-completion, formatting
- **Integrated Terminal:** Run commands and scripts inside the editor
- **Git Integration:** Built-in version control for commits, branches, and merges
- **Debugging:** Set breakpoints, inspect variables, and step through code
- **Extensions:** Thousands of plugins for languages, frameworks, themes
- **IntelliSense:** Smart code suggestions and parameter hints
- **Live Share:** Real-time collaborative coding with others
- **Task Runner:** Automate building, testing, and deployment tasks
- **Snippets:** Reuse common code patterns easily
- **Customizable UI:** Themes, icons, shortcuts, and personalized layout

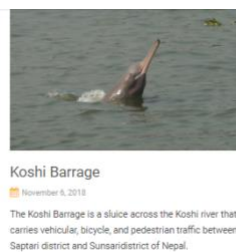
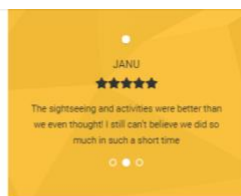
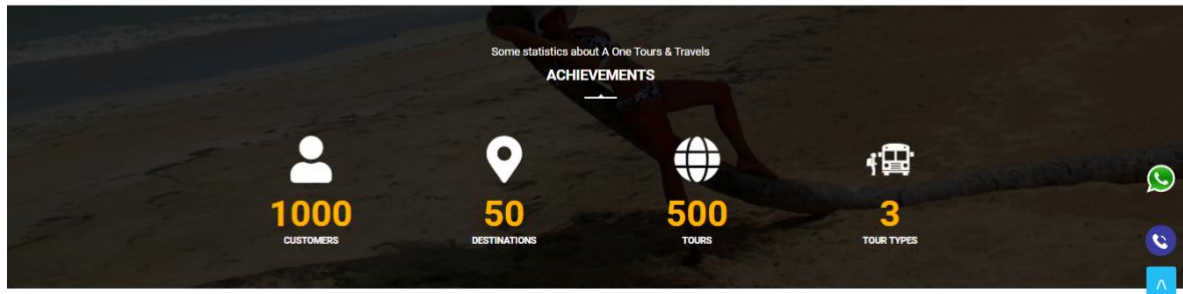
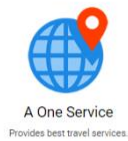
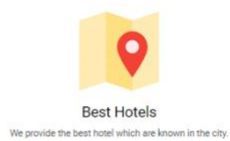
CHAPTER 7

IMPLEMENTATION RESULTS

1. Home Screen :Home page showing the title and welcome message .



WHY CHOOSE US?



CONTACT

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+977-9640990838, 025580923
aoneapp@gmail.com

OTHER SERVICES

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Train ticketing
All Top up

OUR MENU

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- Destinations
- Tours
- Contact

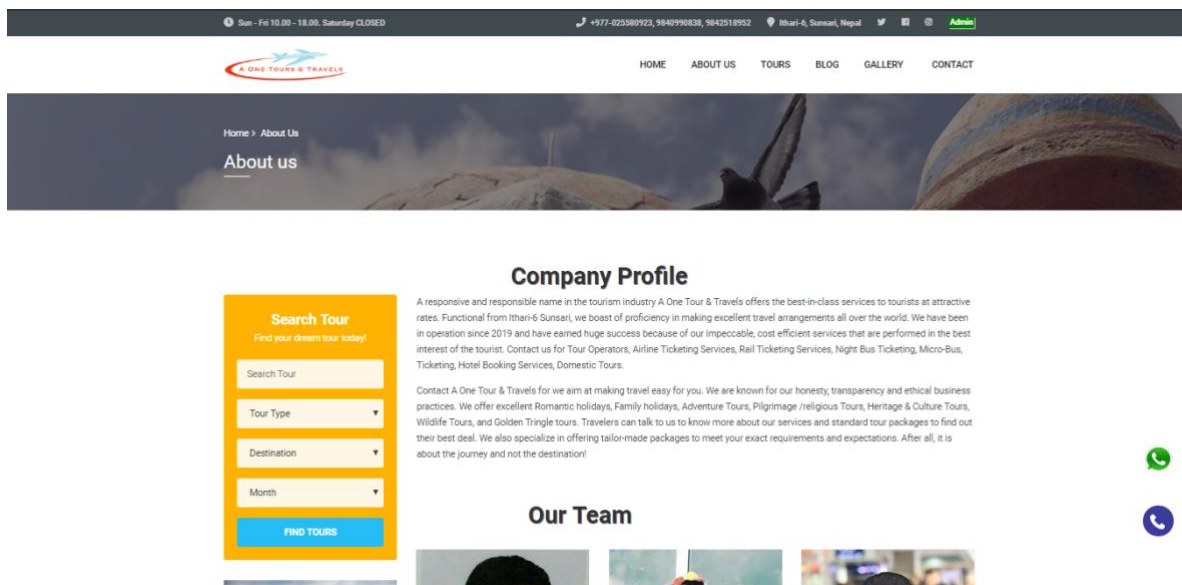
INSTAGRAM

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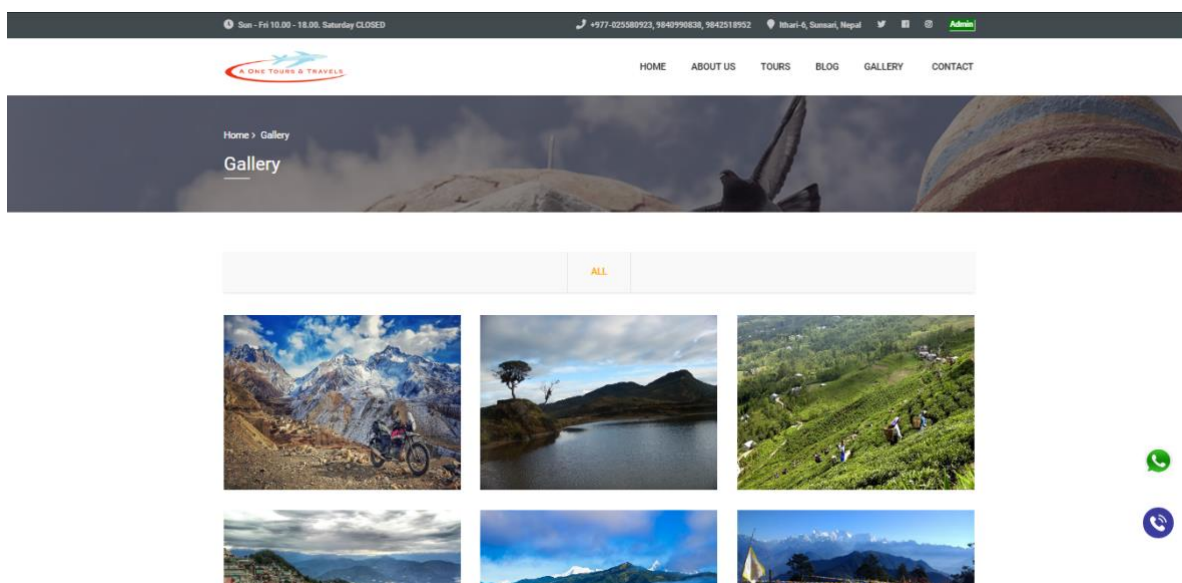
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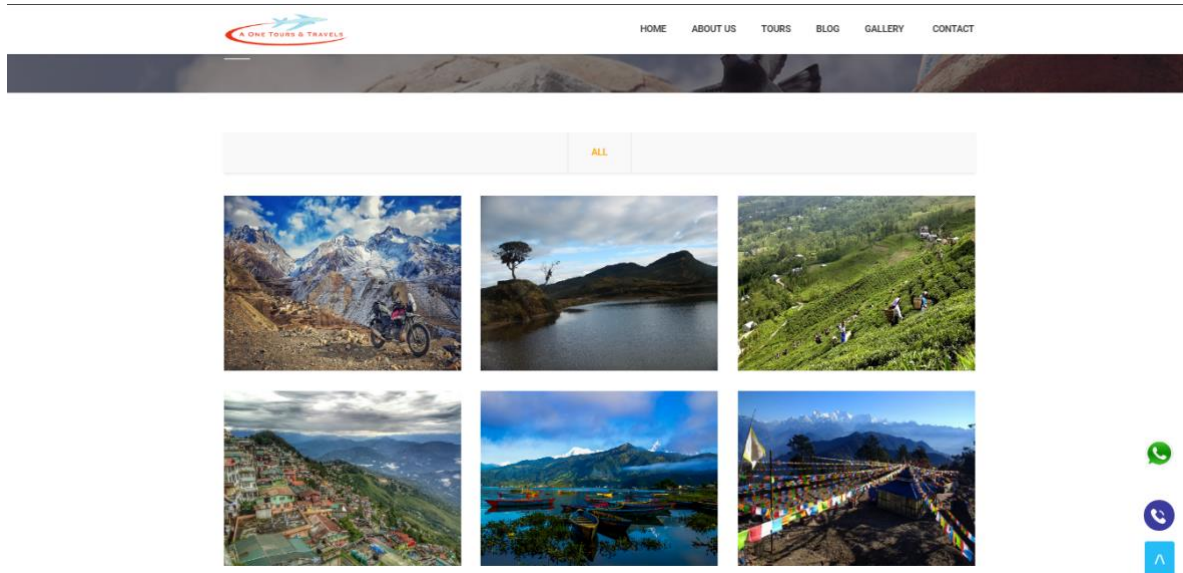
TRAVEL MANAGEMENT

2.About us: Explains who we are, what we do, and the purpose of our organization in a short and clear way

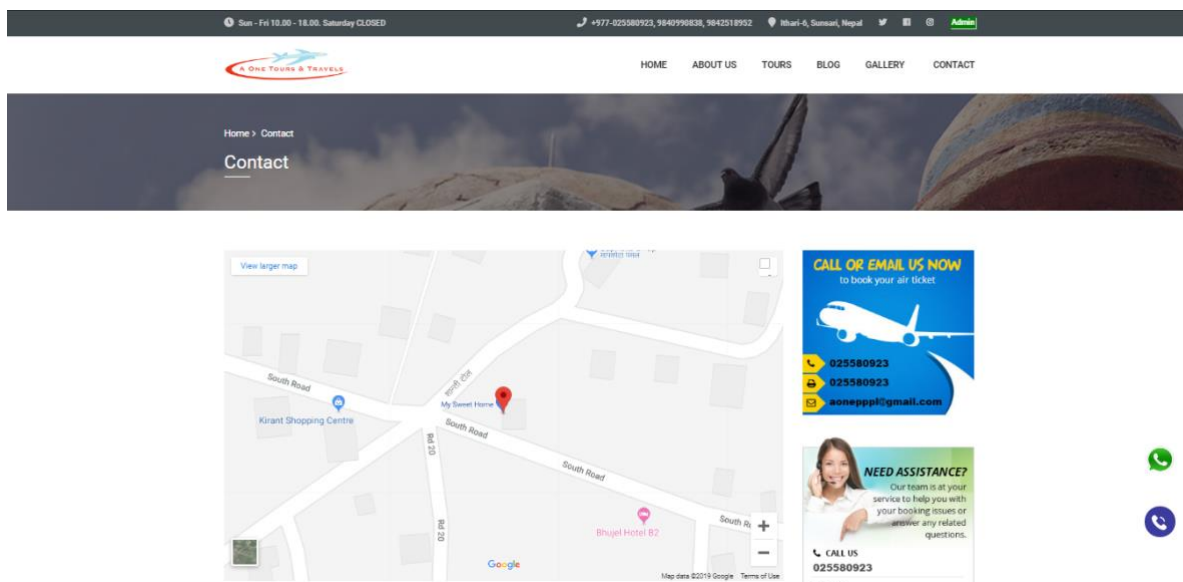


3.Gallery:





4.Contact us:





CHAPTER 8

SYSTEM TESTING

8.1 Introduction to Testing

Testing is a crucial part of the software and data science lifecycle. It ensures that the system performs correctly, handles data properly, and meets the intended requirements. In this project, testing focused on verifying:

- Correct functioning of data preprocessing and analysis modules
- Accuracy of statistical calculations
- Valid and meaningful visual outputs
- System robustness against missing, invalid, or inconsistent data.

8.1.1 Types of Testing:

Unit testing:

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing:

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test:

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

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Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System:testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing:

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing:

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.

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- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format.
- No duplicate entries should be allowed.
- All links should take the user to the correct page.

Integration Testing:

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results:

All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing:

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results:

All the test cases mentioned above passed successfully. No defects encountered.

CHAPTER 9 CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

- Tour and Travel Management System simplifies the management process in travelling.
- Fast processing and immediate results with high security.
- Minimizing human effort and cost efficient database.
- Navigation through the site is easy.
- Any tourist agency can make use of it for saving customer details in database.
- Tourism group can use it for managing their location, hotel, vehicles details.
- This application can easily implement under various situations.
- Reusability of this application is also possible.

9.2 Future Scope:

This is a well-defined job of our team that has taken six months to analyze situation in which this project will work and the environment of the company that will use this project. It was a good experience for us to work in a company that has very friendly environment and learning atmosphere.

Who gave a lot of good programming skills and so many company related skills that will help me in the future. I am also thankful to all my friends and team members.

9.2.1 Software Scope:

Reusability:

Reusability is possible as and when we require in this application. We can update it next version. Reusable software reduces design, coding and testing cost by amortizing effort over several designs. Reducing the amount of code also simplifies understanding, which increases the likelihood that the code is correct. We follow up both types of reusability: Sharing of newly written code within a project and reuse of previously written code on new projects.

Extensibility:

This software is extended in ways that its original developers may not expect. The following principles enhance extensibility like Hide data structure, avoid traversing multiple links or methods, avoid case statements on object type and distinguish public and private operations. This software is extended in ways that its original developers may not expect. The following principles enhance extensibility like Hide data structure, avoid traversing multiple links or methods, avoid case statements on object type and distinguish public and private operations.

Robustness:

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Its method is robust if it does not fail even if it receives improper parameters. There are some facilities like Protect against errors, optimize after the program runs, validating arguments and Avoid predefined limits.

Understandability:

A method is understandable if someone other than the creator of the method can understand the code (as well as the creator after a time lapse). We use the method, which small and coherent helps to accomplish this code also simplifies understanding, which increases the likelihood that the code is correct. We follow up both types of reusability: Sharing of newly written code within a project and reuse of previously written code on new projects.

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